Attention · C3

Rank eligible work.
Credit each cause once.

ELI16 · THE IDEA IN PLAIN LANGUAGE

A priority list with a clock: tasks gain or lose points, but a repeated reward receipt cannot buy a second reward.

Why it matters: Limited work budgets make ordering observable. Keeping attention independent allows policy/provider experiments without changing belief mathematics or task admission.

Through M5 · 17 October 2026 · 12–15 minute talk · evidence and proposals kept separate

One module in a system of fifteen

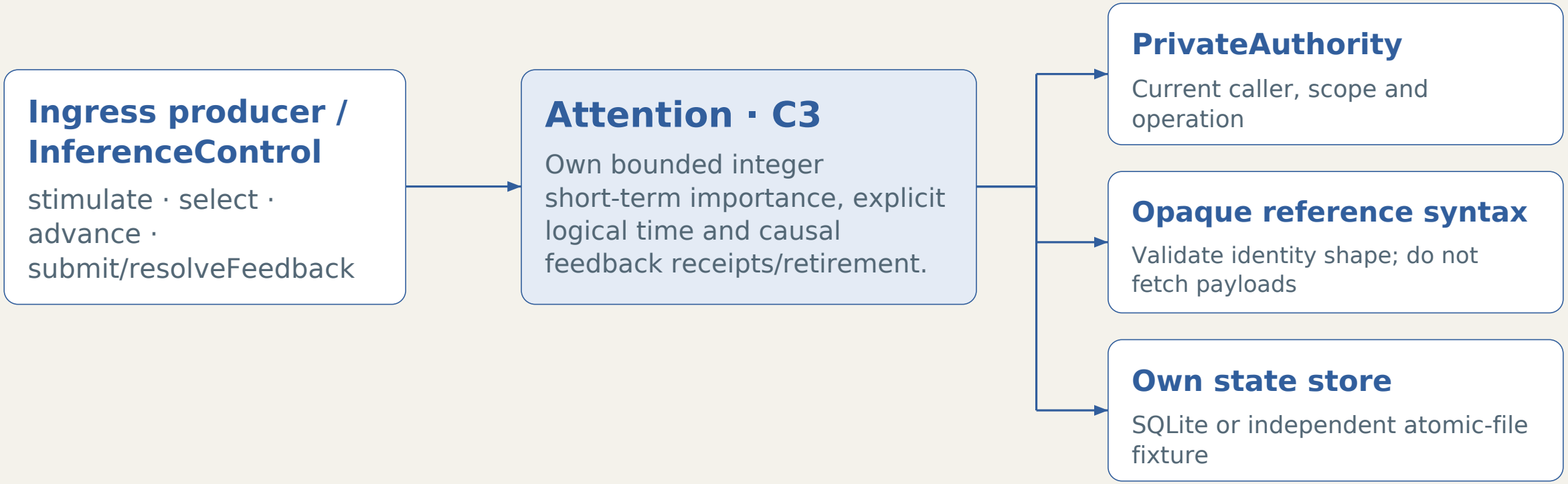
The highlighted responsibility is the subject of this talk; placement and process count are separate decisions.

Persistence	Knowledge access	Cognition	Coordination + policy
DA storage	Relations	Episodes · C1	Messaging
AtomStorage · C0	Search projection	Beliefs · C2	Execution
ConsensusFinality		PLNKernel	Registry
		Attention · C3	Resolver
		Inference · C4	Authorization

Own bounded integer short-term importance, explicit logical time and causal feedback receipts/retirement.

Dependencies explain the boundary

Arrows point from caller to dependency. Stores and runtime internals are labelled explicitly.



One responsibility does not imply one process. A dependency does not grant access to its private tables.

Responsibility, state and scope

The owner is authoritative for its own state; other responsibilities are deliberately separate.

OWNS / GUARANTEES

Priority and logical clock

STI per target/context, revision and generation watermark

Causal outcome ledger

Command replay, feedback application and no-credit retirement

OUTSIDE THIS RESPONSIBILITY

Truth values

STI is not belief confidence and does not alter formula inputs

Cognition/search

No task creation, ECAN, LTI, wages/funds or hidden inference dependency

sti-additive-linear-decay/1; fresh closure profile sti-additive-linear-decay-closure/1

The executable surface

These operations and implementation seams define the supported scope; versions and profiles remain explicit.

OPERATION / SEAM	RESPONSIBILITY	CONTRACT CONSEQUENCE
submitStimulus / readAttention	Ingress plus explicit read	Reads do not allocate or decay state
select	Rank positive eligible STI; deterministic tie-break	No discovery of new tasks
advance	Explicit expected-step logical decay	Stale clocks fail; replay does not retime
submitFeedback / resolveFeedback	Apply causal credit or account for prior outcome	Closure cannot newly apply eligible feedback

Default 16 targets/context, 8 contexts; finite cause/command capacity; explicit logical steps.

A worked example through the boundary

HISTORICAL M3 CONTROLLED COMPARISON · SQLite + independent file fixture

01 · STEP

Prefer B

Scores A=10, B=20; one attempt available.

02 · STEP

Select

B runs first; C2 admits one conclusion.

03 · STEP

Credit once

C3 records one causal feedback outcome.

04 · STEP

Replay / swap

Both providers retain outcome; no extra native call.

What the example establishes

Priority selects work; causal receipts prevent repeated reward.

Four invariants to preserve

Each statement is bounded by the selected profile, authority model and tested failure model.

I1

Policy is explicit

Integer saturation, context budget and decay are part of identity.

I2

Causal uniqueness

Replay cannot apply the same logical feedback twice.

I3

Strict logical time

Feedback is never retimed to make stale work admissible.

I4

One authoritative outcome

Application and retirement serialize at the attention owner.

Failure is part of the interface

A lost response, a missing prerequisite and a deliberate refusal require different interpretations.

BOUNDARY / TRIGGER	OWNER-AUTHORITATIVE OUTCOME	EVIDENCE / LIMITATION
Feedback applied; response lost	Return original application receipt even after expiry/decay.	No second credit
Clock advanced / feedback expired	Retire without credit when permitted.	Do not manufacture application
Eligible/unavailable or finite capacity	Resolution remains pending.	Reconciliation is not a scheduler

Recovery evidence covers same-host process termination with intact files; it does not prove power-loss durability.

Measurements and benchmark gaps

Observed values, contract limits and proposed experiments are explicitly different kinds of evidence.

4

Controlled provider/preference runs

Same policy, tasks and one-attempt budget

1 / 0

Native call / replay calls per run

Whole inference-step observation

100 / 300 / 1

Target cap / context budget / decay

Default STI policy parameters

Historical full-step timings span 0.641150–0.744361 s, one observation per case. They include native inference and cannot rank attention-provider speed or cognitive quality.

PROPOSED BENCHMARK · NOT RUN

Proposed: fix eligible sets and causal schedule; vary target count, replay ratio and feedback contention. Measure select/update p50/p95 while checking exact ranking and once-only outcomes.

Current state: what is real, what is not

Current supported regressions pass; historical observations and unsatisfied capabilities keep their labels.

CLASSIFICATION	MODULE-SPECIFIC STATE
REAL / IMPLEMENTED	Persistent SQLite reference STI provider; real native inference consumes its selections.
FIXTURE / LOCAL TEST POLICY	Independent atomic-file attention, generated tasks, grants and deterministic fault positions.
BLOCKED / UNSUPPORTED	No required supported gate blocked; old pending stores need their original profile.
UNRUN	ECAN/LTI, learned policy quality, production scaling and populated-state migration.

M5 final tested runtime: bfb0ec8 · evidence: 3dc6c11 · module representation is not product completeness.

Future directions need acceptance criteria

These are discussion options, not a committed roadmap or permission to expand the current profile.

DIRECTION	PREREQUISITE / DESIGN GATE	MEASURE OF SUCCESS
Benchmark independent providers	Identical logical schedule and durability semantics	Performance without ranking or receipt drift
Evaluate ranking usefulness	Declared task objective and fixed inference budget	Outcome quality separate from conformance
Qualify another policy	New profile with clock/causal semantics	No hidden adoption of LTI or coupled attention state

Keep ownership, current authorization and explicit compatibility intact; qualify new semantics with a new profile.

Evidence you can inspect and reproduce

Read the exact code, contracts and recorded run before treating a claim as a guarantee.

Takeaway

Priority selects work; causal receipts prevent repeated reward.

SUPPORTED REPRODUCTION ENTRYPOINT · PREREQUISITES REQUIRED

```
uv run --python 3.11 --locked python scripts/verify_m3_1.py <env> --output <fresh>
```

Uses fresh closure-profile stores and both providers. Historical comparison retains its own source label; M5 reruns legacy behavior without adding attention work.

`fusion/attention.py`
`fusion/attention_contracts.py`
`fusion/adapters/fixture_attention.py`
`tests/test_m3_1_attention.py`

Source 494df69a1ac401c3d3b2cf79dbfde07f74b5e277 · all citations and full notes accompany this deck.